

Henil Parmar | Curriculum Vitae

Student at IIT Bombay major in Aerospace Engineering

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EDUCATION

Indian Institute of Technology Bombay

[2023–2027]

Bachelor of Technology in Aerospace Engineering

CPI: 7.42 / 10 (Ongoing)

PROFESSIONAL EXPERIENCE

Airbus Defence and Space | Aerospace Engineering Intern

[Jun '26–Jul '26]

C295 Make in India Program, Final Assembly Line, Vadodara

- Analyzed **C295** Final Assembly Line workflows, mastering station-wise operations & complex system integration in context
- Validated critical assembly quality across multiple production stages utilizing advanced **aerospace test equipment** onsite
- Identified process bottlenecks & learned about **lean manufacturing solutions** for optimization of overall FAL productivity
- Streamlined large-scale assembly operations via **cross-functional coordination** across **Engineering and Quality** teams

Indian Space Research Organisation (ISRO) | Research Intern

[May '25–Jul '25]

Reusable Launch Vehicle guidance-algorithm development under Dr. Ashish Kumar Shukla

- Developed an **AI-powered** guidance algorithm using **GPS & INS** for precise **RLV autonomous landings** with accuracy
- Optimized **Pseudolite navigation** via **Monte Carlo simulations** to enhance robustness of guidance algorithm for RLV
- Developed a **multi-modal guidance algorithm** establishing precise, true autonomous landing capabilities for the **RLV**
- Integrated mathematical models into visual simulations, presenting **RLV control advancements** to senior scientists well.

RESEARCH EXPERIENCE

Reusable Launch Vehicle (RLV) Research Project | IIT Bombay

[Sep '24–Nov '25]

Aerospace Vehicle Design Series (~1.5 years) | Guide: Prof. Dhwanil Shukla

- Investigated autonomous **RLV return-and-landing** via mission trajectory and landing-system design with prof. guidance.
- Developed **G-FOLD & terminal-velocity** guidance for sub-2 m/s controlled descent of a 549-tonne LEO Launch Vehicle
- Analyzed **7.2% return fuel allocation** for a **2,858 m/s Δv target** linking guidance targets to landing-system constraints.
- Designed four landing legs with **CAD** for a **472.4 kN touchdown impact load**, linking structural and guidance constraints.

KEY PROJECTS

F1 Minimum-Lap-Time Simulator | Self Project

Vehicle dynamics, optimal control and telemetry-based validation

- Developed a Python minimum-lap-time optimal-control simulator using **CasADi/IPOPT** and degree-three **Radau direct collocation**, jointly optimizing racing line and speed on six real F1 circuit centerlines, for realistic performance analysis.
- Implemented configurable open-wheel vehicle physics with quadratic drag/downforce, power- and tractive-force-limited acceleration, quasi-static load transfer, and **load-sensitive combined longitudinal/lateral tyre-force constraints** used.
- Integrated **FastF1 telemetry comparison** for lap-time error, speed RMSE, cumulative time delta, sectors/corners and braking zones; benchmarked in 2021 Monza at **+6.28% lap-time error** and **24.29 km/h speed RMSE** without calibration

Energy Management and Control | Self Project

[Ongoing]

Dynamic modelling, control-strategy development and constrained optimization

- Developing a system-level energy-management model to study power/energy flow and formulate control objectives, operating constraints, and supervisory control logic.
- Formulating an optimization-oriented control strategy to allocate energy across competing system demands; evaluating controller behaviour and trade-offs through simulation as the model is developed.

Satellite Attitude Control using LQR | Course Project

[Aug'26– Present]

State Space Methods for Flight Vehicles | Guide: Prof. Shashi Ranjan Kumar

- Modelled 3-axis satellite attitude dynamics as a 6-state **state-space model**, linearized around nominal equilibrium plane
- Analyzed **controllability and observability**, then designed an **LQR state-feedback law** for satellite attitude regulation
- Simulated closed-loop satellite response and tuned **LQR weights** to improve settling, tracking accuracy and control effort

STOL Aircraft & Airfoil Design | Course Project

[Jan'24– Apr'24]

Low Speed Aerodynamics | Guide: Prof. Dhwanil Shukla

- Built CFD meshes in **ANSYS** to model 2D subsonic flow over a **NACA 5311**, with peak C_l/C_d of **61.11** at 15° stall angle.
- Made python tool with 2 peers for computing C_l , circulation & induced velocities using Thin airfoil theory for custom airfoil
- Built 3D STOL aircraft aerodynamics in **OpenVSP** via empirical drag methods for **157 m** takeoff roll, boosting C_l by **8.3%**

Sensors, Actuators and MEMS FEA

Finite-element modelling of sensing/actuation structures

- Modelled MEMS sensing/actuation structures using **finite-element analysis**, examining structural/device response under relevant loading and design-parameter variations.
- Studied design sensitivity and sensing/actuation behaviour to connect device-level mechanics with system-level measurement and control requirements.

Universal Testing Machine | Course Project

[Aug'23–Nov'23]

Makerspace | Guides: Prof. Joseph John, Prof. K.N. Jonnalagadda

- Engineered and built **Universal Testing Machine (UTM)** to experimentally measure **Young's modulus** of test specimen
- Designed & built **lead-screw transmission system** to apply **controlled mechanical loads** across material test samples
- Integrated force and displacement tracking using **Arduino UNO**, **IR sensor PWM** to automate test data curve generation

Sensor Fusion for Human Motion State Estimation | Course Project

[Jan'26 – Apr'26]

Joint Biomechanics | Guide: Prof. Darshan Shah

- Built a **MediaPipe Pose** 2D vision pipeline to extract hip, knee, ankle keypoints from 24 images across 6 test subjects
- Processed pose and accelerometer data across 6 subjects and 4 static poses, achieving $r = 0.984$ correlation in angles
- Validated markerless pose estimates against goniometer data, achieving $r = 0.990$ agreement for **computed moments**

TECHNICAL SKILLS & COURSES

Programming / Computing	Python, C++, MATLAB, MySQL, HTML, $\LaTeX$
Scientific Libraries	NumPy, Pandas, Matplotlib, SciPy
Optimization / Simulation	CasADi, IPOPT, ANSYS, OpenVSP
CAD / Engineering Tools	Fusion 360, SolidWorks, LaserCAD, Arduino IDE
Controls / Aerospace	Control Theory; Flight Mechanics of Aircraft and Spacecraft; Aircraft Dynamics;
Coursework	Aerodynamics; Aero Propulsion; Aircraft Design

LEADERSHIP, AWARDS & SERVICE

Joint Secretary | Aerospace Engineering Department, IIT Bombay

[Apr '25–Apr '26]

Academic coordination, departmental events, and student–faculty engagement

- Formed and led an **8-member council** supporting events, workshops, academic coordination, and faculty engagement for **450+ students**.
- Contributed to development of the department website and digital outreach activities.

Student Mentor | Department Academic Mentor Program, IIT Bombay

[Jul '25–Present]

Mentor for second-year students

- Selected as one of **28 mentors** through a competitive process; provided regular academic and social guidance to second-year students.
- Awarded **Best DAMP Mentor** among 28 peers for academic guidance and student support in 2026.

System Administrator | Hostel 16 Council, IIT Bombay

[Dec '23–May '24]

Network and computing support for hostel residents

- Helped deploy IITB Wireless and improve LAN/network services for approximately **1,000+ students**, in coordination with IIT Bombay's Computer Centre.

Additional service: Academic Coach at Vidhyadaan, supporting **20+ youth** through literacy and learning initiatives; NSS volunteer supporting underprivileged students.

Interests: Football, basketball, hiking and forts, board games, video games, and science-fiction reading.